

LLM Agentic Workflow for Automated Vulnerability Detection and Remediation in Infrastructure-as-Code

127003056 Dhanvanth.V
127003252 Someshwar.S

Abstract:

Problem Statement:

The rapid adoption of Infrastructure-as-Code (IaC) in cloud-native environments has introduced significant security challenges, where misconfigurations in declarative templates can lead to severe data breaches. As cloud infrastructures grow in complexity, identifying subtle, context-specific vulnerabilities becomes increasingly difficult for developers relying on traditional tools.

Existing Methodologies:

Current vulnerability detection methods primarily consist of static analysis tools, rigid rule-based systems (such as CDK-Nag), and dynamic runtime analysis. While effective for known issues, these methodologies often fail to detect interdependent vulnerabilities and generate high rates of false positives due to their inability to interpret the specific context of the infrastructure. Furthermore, they rarely provide automated, context-aware remediation guidance, impeding efficient security scaling in CI/CD pipelines.

Proposed Solution & Results:

This project proposes a multi-agent, AI-driven strategy employing Large Language Models (LLMs) and Retrieval-Augmented Generation (RAG) to detect and remediate security flaws. By utilizing a continuously updated knowledge base of best practices, the system analyzes IaC templates to discern context-specific misconfigurations. Experimental results demonstrate that this approach achieves a detection rate of 85% for complex vulnerabilities and provides actionable, automated remediation steps that integrate seamlessly into standard development workflows, thereby reducing manual intervention and enhancing overall cloud security.

References:

[1]. [Dheer Toprani and V. K. Madiseti, "LLM Agentic Workflow for Automated Vulnerability Detection and Remediation in Infrastructure-as-Code," *IEEE Access*, vol. 13, pp. 69175–69181, 2025](#)

Guide Name: Dr.R.Alageswaran